

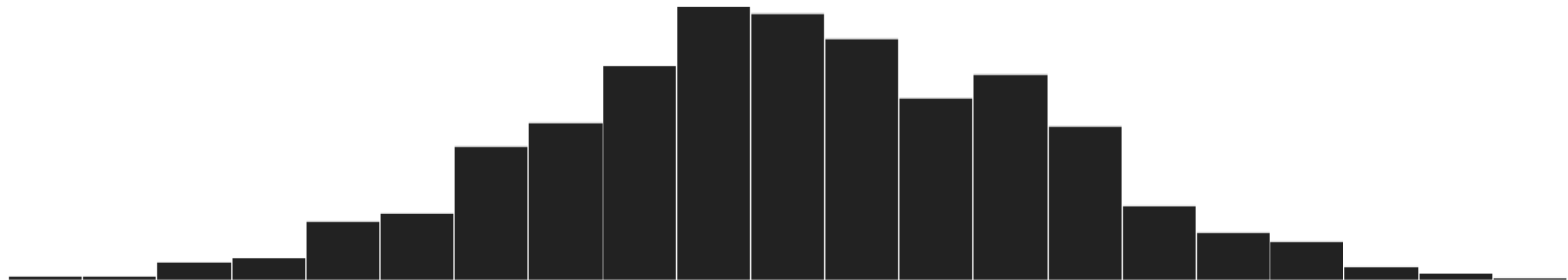
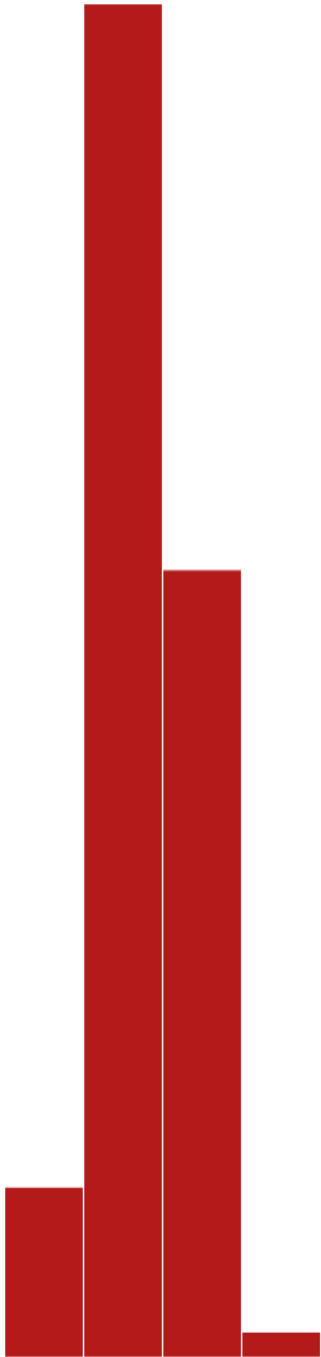
VTPEH 6105

Lecture 05

HYPOTHESIS TESTING: COMPARISON OF MEANS ACROSS MULTIPLE GROUPS

Amandine Gamble

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House Keeping

Feedback on Previous Homework

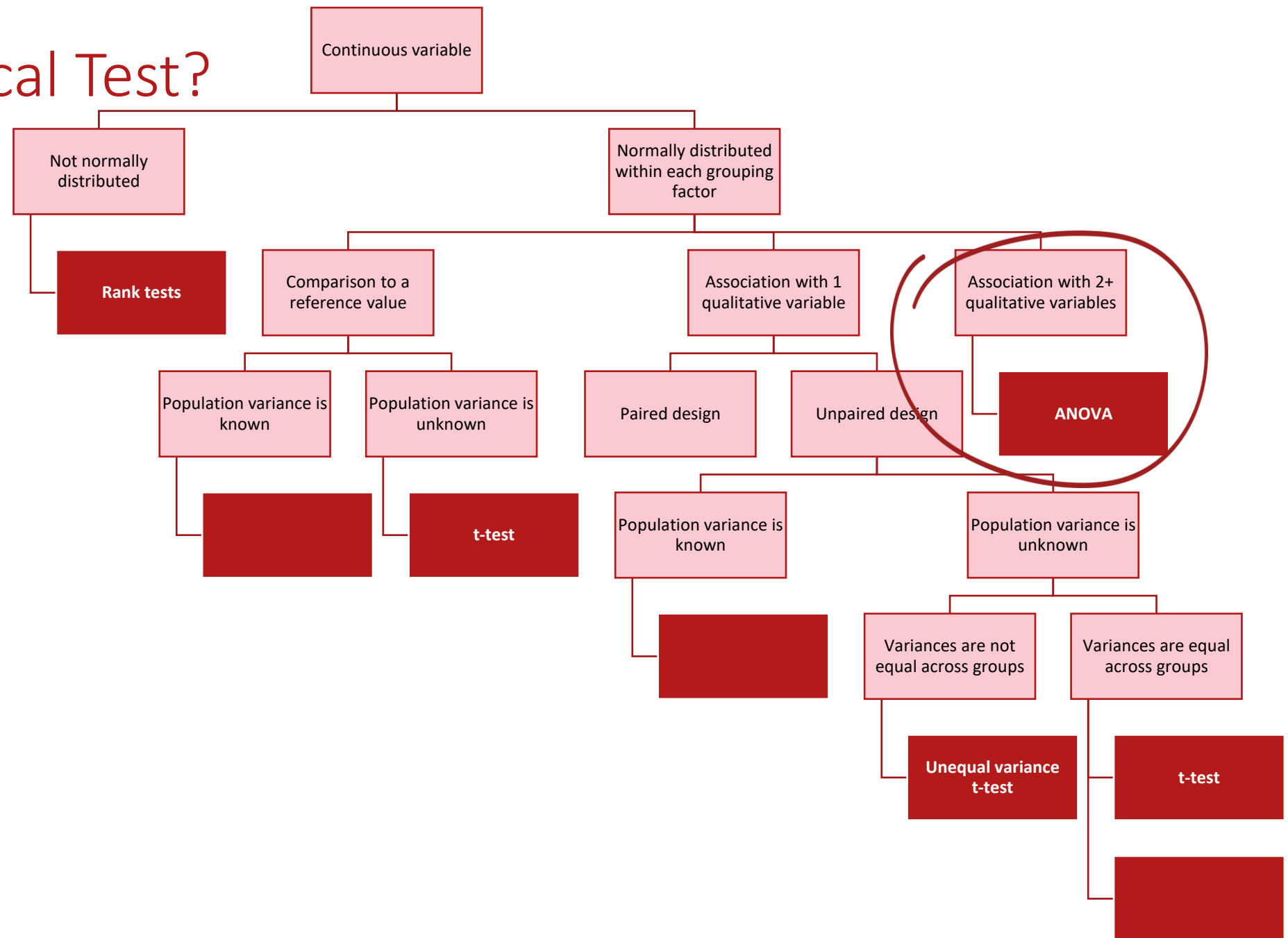
- Good overall (=
- Pay attention to formatting, proof-read the PDF...
- Questions all referred to Sharp et al. 1996

Hypothesis Testing Refresher

Which Hypotheses?

- H_0 : there is **no association** between the variables X and Y (knowing X does not inform on Y)
- H_1 : there is an association between the variables X and Y (knowing X at least partially inform on Y)
- **p -value** is the probability of **observing the data under the assumption that H_0 is true**
 - If p is very low: unlikely to observe the data if H_0 were true, but we do observe the data, so we reject $H_0 \rightarrow$ There is a significant association between X and Y
 - If p is high: likely to observe the data if H_0 were true, and we do observe the data, so we do not reject $H_0 \rightarrow$ There is no significant association between X and Y

Which Statistical Test?



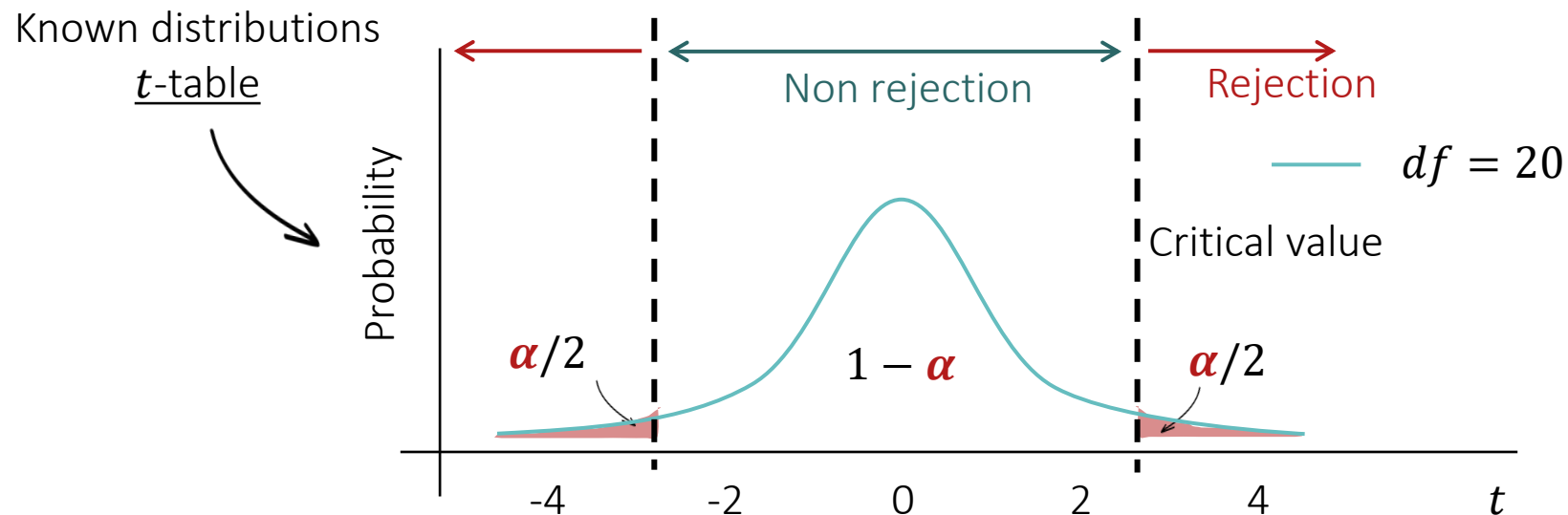
Statistical Tests: the T-test, How?

- Two-sample: 2 means together (μ_1 and μ_2)

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2 \frac{1}{n_1} + \frac{1}{n_2}}}, df = n_1 + n_2 - 2$$

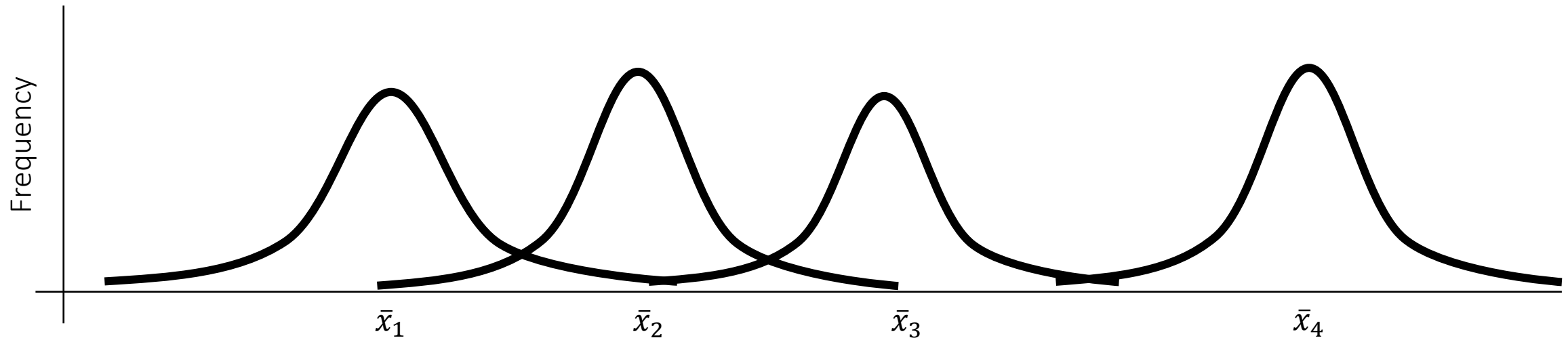
With $\bar{x}$ the sample mean, μ_0 the reference value, s the sample standard deviation and n the sample size

With $\mu_1 - \mu_2 = 0$ under H_0 , and $\bar{x}_1 - \bar{x}_2$ close to 0, hence z close to 0 if H_0 is true



Comparison of Means

When there is more than 2 groups... Analyses of variance, or ANOVAs



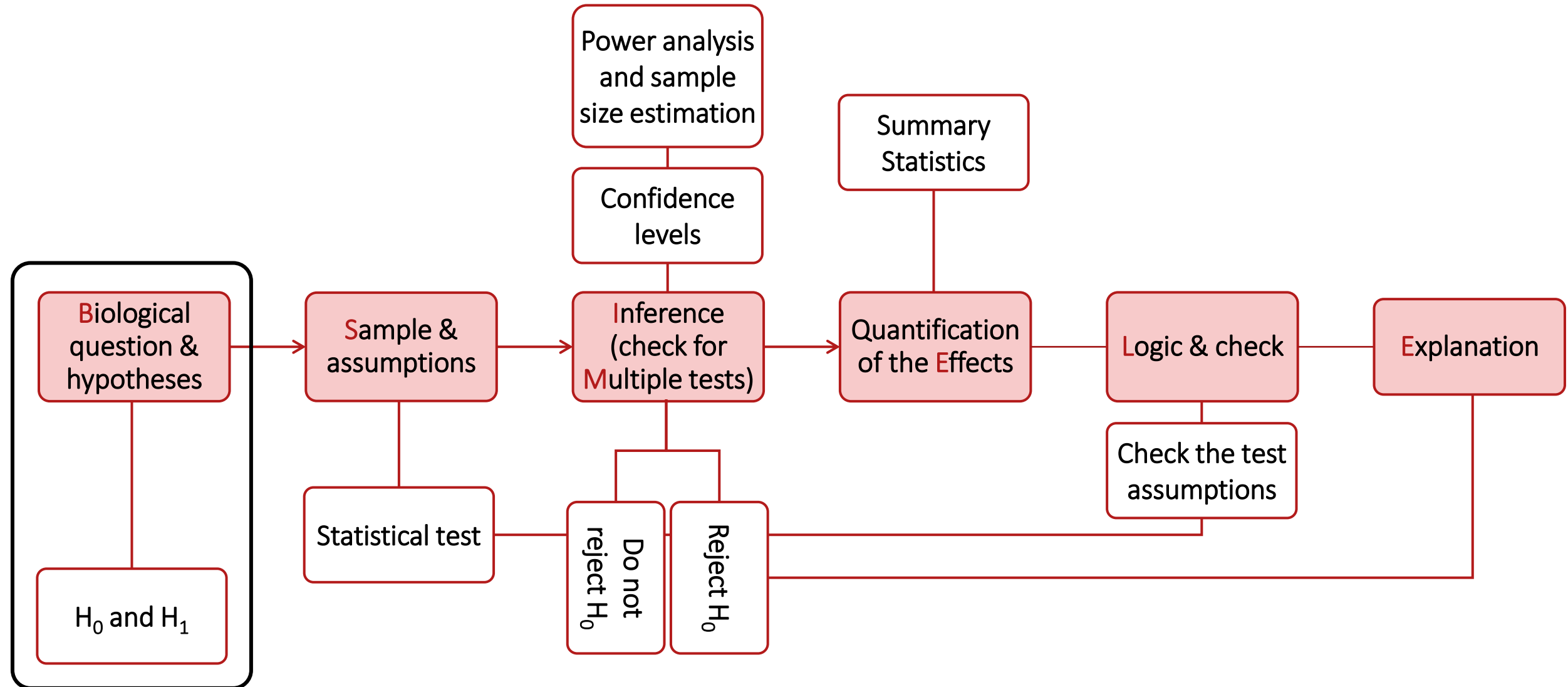


Hypothesis Testing Considerations

B-SIMPLE

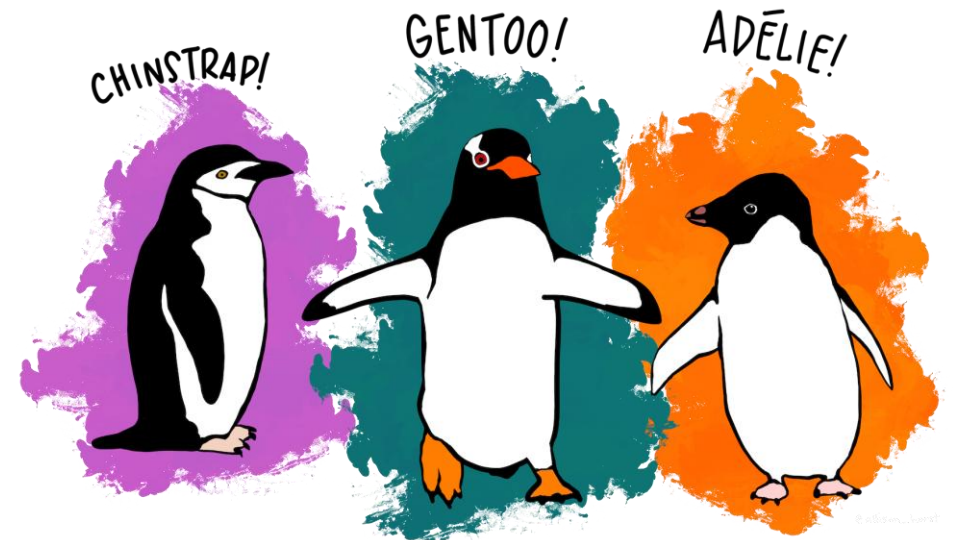
- **Biological question and research hypothesis** What am I actually trying to answer?
- **Sample and assumptions** Which approach can I reasonably use?
- **Inference power and significance** Can we, and do we, detect an association?
- **Multiple comparisons and adjustments** Do the thresholds need to be adjusted for multiple testing?
- **Practical effect** What is the magnitude of the association?
- **Logic and check** Does the output make sense? Are the test assumptions verified?
- **Explanation** How do I communicate the results to my audience?

Hypothesis Testing Workflow

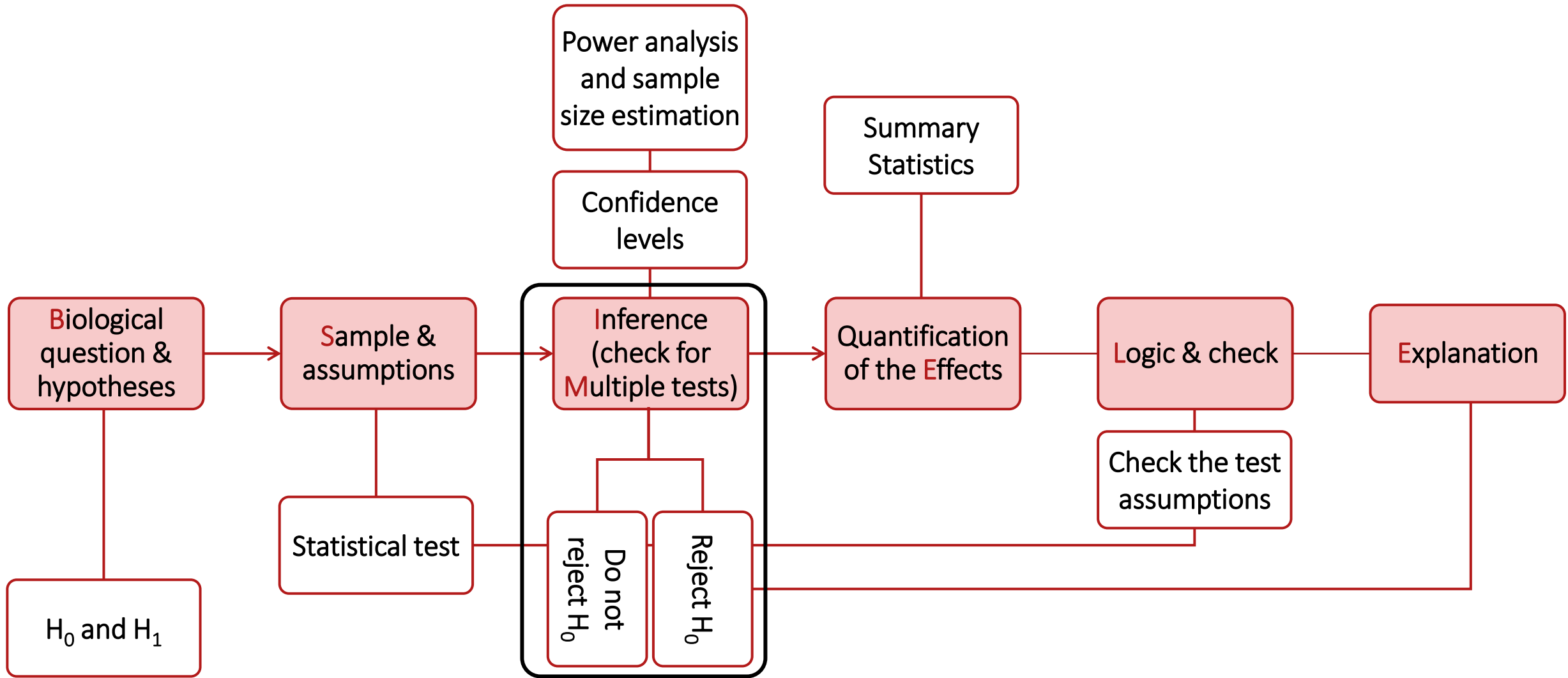


Working hypotheses

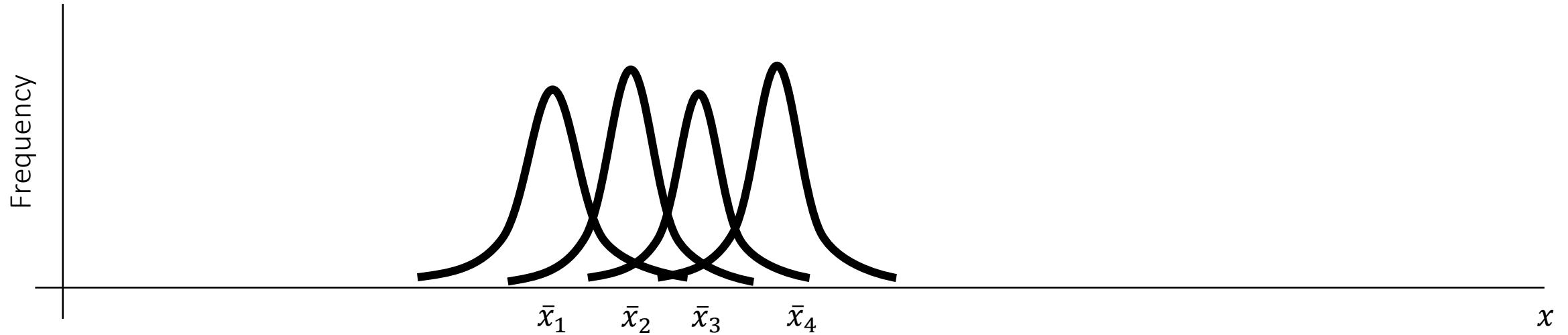
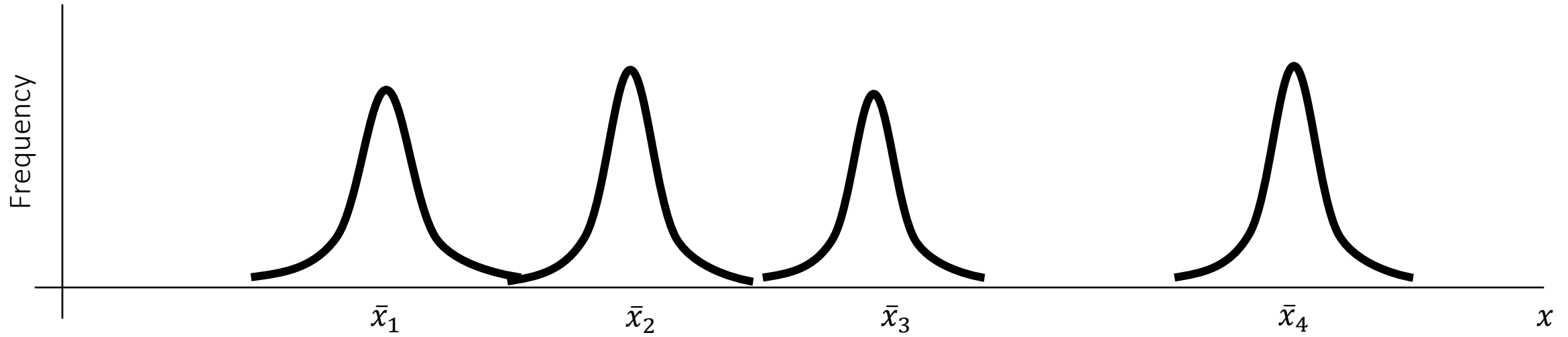
- Null hypothesis (H_0): $\mu_1 = \mu_2 = \mu_3$
 - Calory intake is no different across countries
 - There is an association between country and calory intake
- Alternative hypothesis (H_1): $\mu_1 \neq \mu_2$ or $\mu_1 \neq \mu_3$ or $\mu_2 \neq \mu_3$
 - Calory intake is different across countries
 - There is no association between country and calory intake



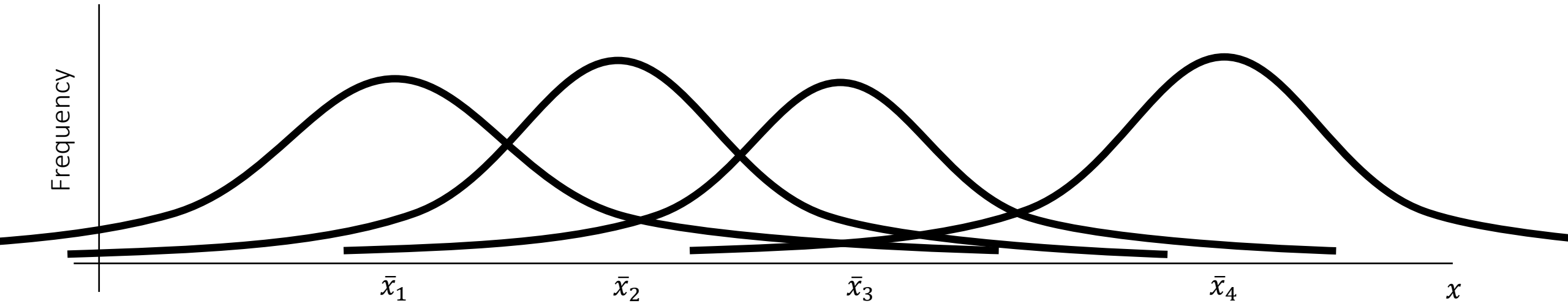
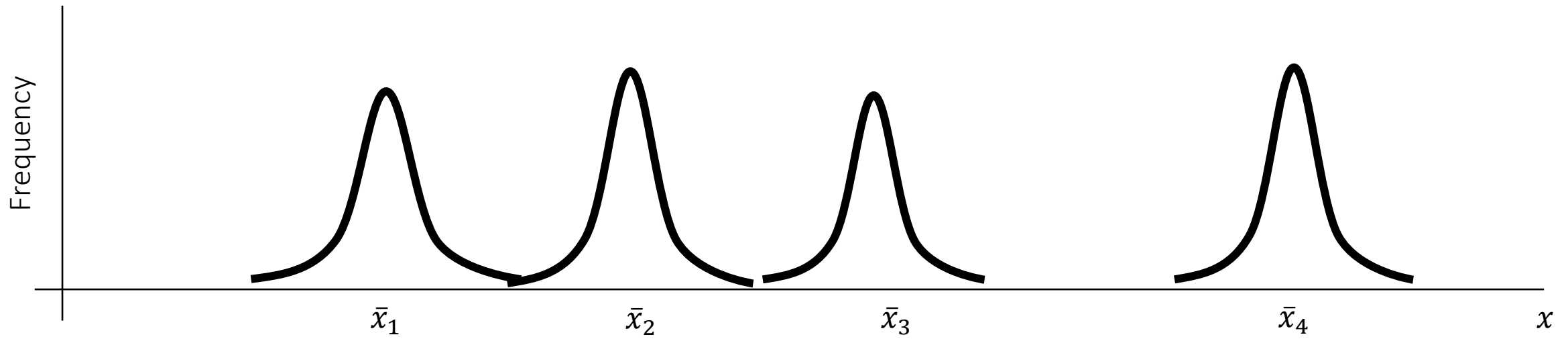
Hypothesis Testing Workflow



ANOVA – How? With intuition...



ANOVA – How? With intuition...

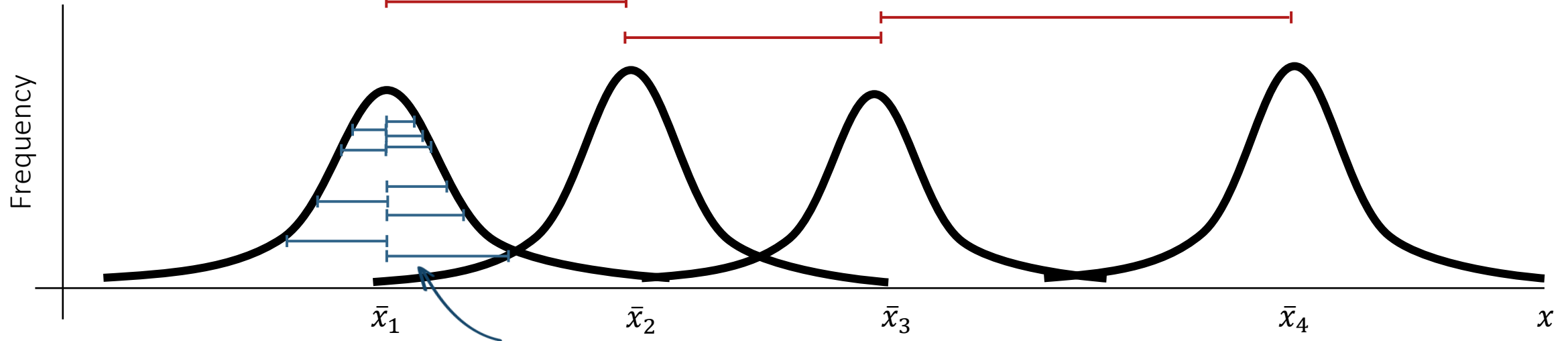
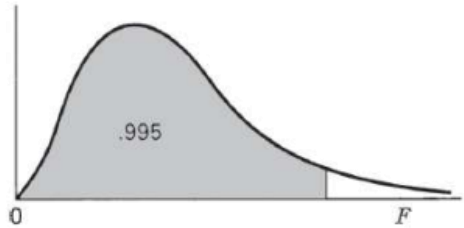


ANOVA – How? With math....

$$F = \frac{\text{Among groups mean square}}{\text{Within groups mean square}}$$

Variation explained by the differences across the groups (i.e., among group variation)

Table G Percentiles of the F Distribution



Residual "noisy" variance (not explained by the groups, i.e., within group variation)

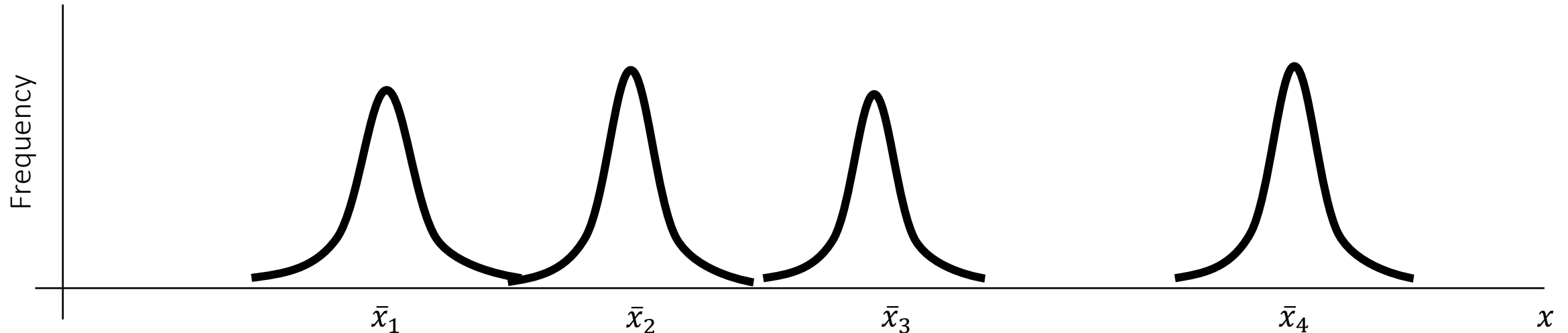
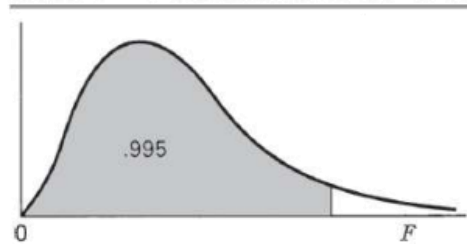
ANOVA – How? With math....

$$F = \frac{\text{Among groups mean square}}{\text{Within groups mean square}}$$

Variation explained by the differences across the groups (i.e., among group variation)

Less likely to observed this F value under H_0

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ANOVA – How? With math....



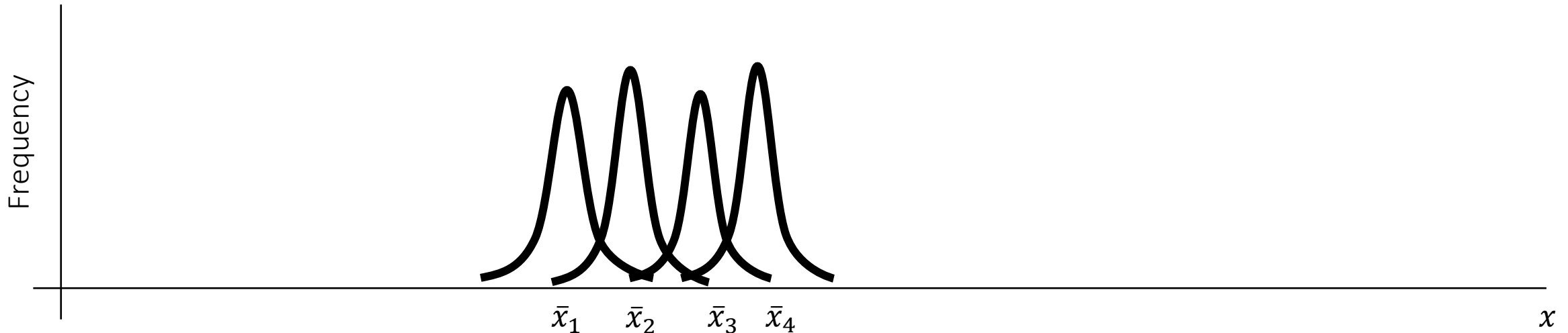
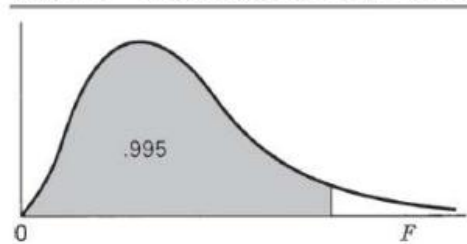
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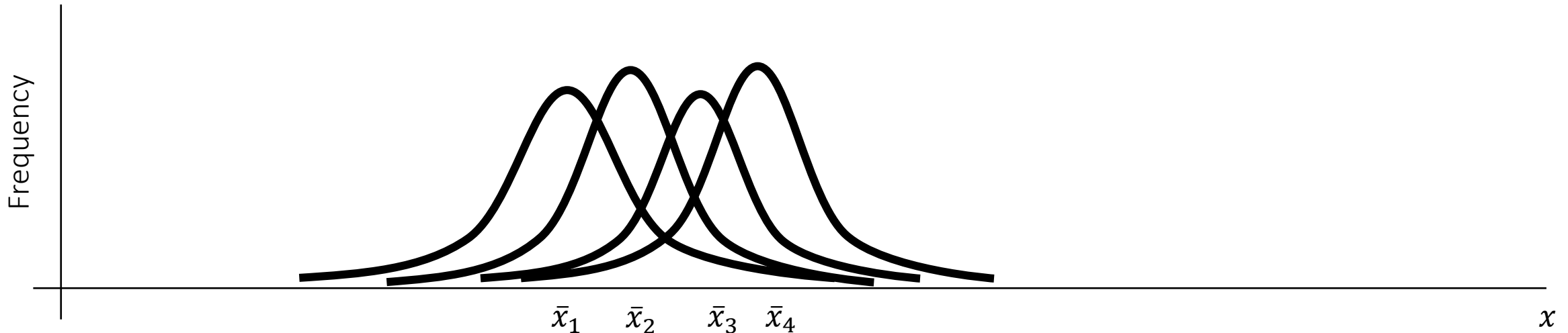
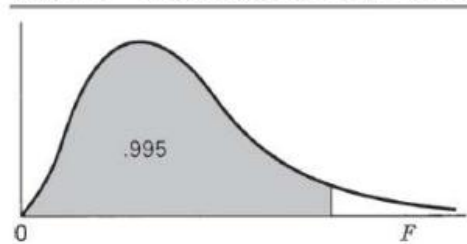
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Residual “noisy” variance (not explained by the groups, i.e., within group variation)

ANOVA – How? With math....

$$F = \frac{\text{Among groups mean square}}{\text{Within groups mean square}}$$

$$F = \frac{MS_{Between}}{MS_{Within}} = \frac{\frac{SS_{Between}}{df_{Between}}}{\frac{SS_{Within}}{df_{Within}}}$$

For each group

$$SS_{Between} = \sum_i \overset{\text{Group size}}{n_i} (\overset{\text{Group mean}}{\bar{Y}_i} - \overset{\text{Overall mean}}{\bar{Y}})^2$$

Number of groups

$$df_{Between} = k - 1$$

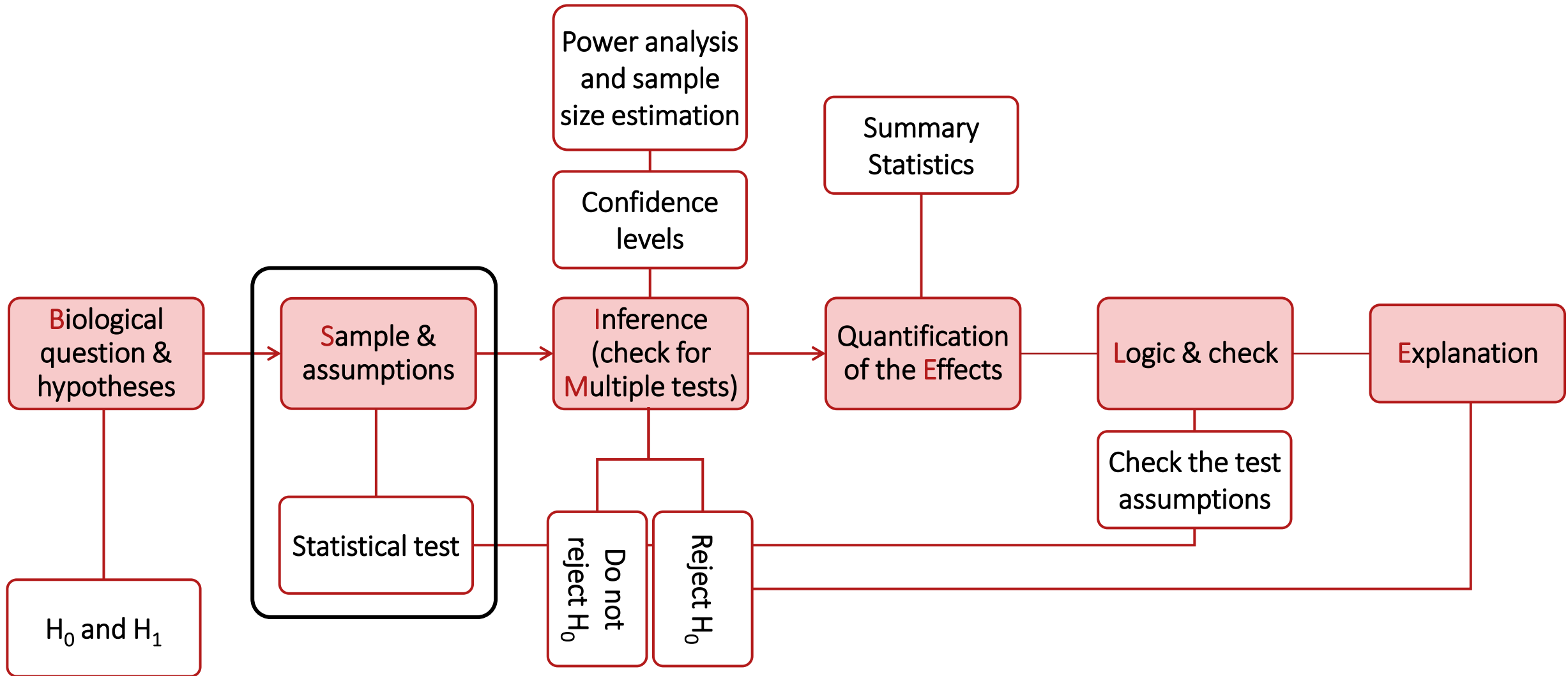
$$SS_{Within} = \sum_i \sum_j \overset{\text{Group mean}}{(\overset{\text{Individual value}}{Y_{ij}} - \bar{Y}_i)^2}$$

For each individual

Number of individuals

$$df_{Within} = N - k$$

Hypothesis Testing Workflow



ANOVA – When? Assumptions

- Each group's samples must be **independent** from the other group
- The populations distributions for each group i must be approximately **normal** mean μ_i and standard deviation σ_i
- Each of the population has an **equal variance** *i.e.*, $\sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2 = \sigma^2$

Study design

Histograms
Shapiro-Wilk test

Relaxed for large sample size but mean often irrelevant for non-normal variables

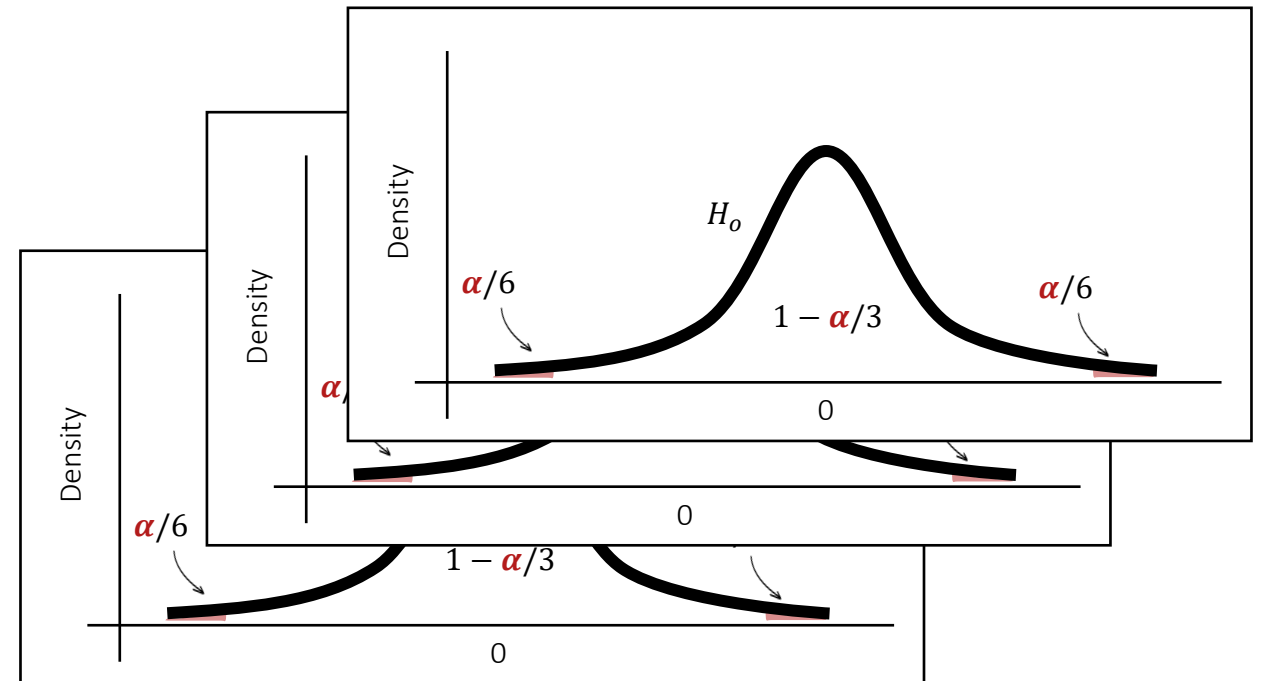
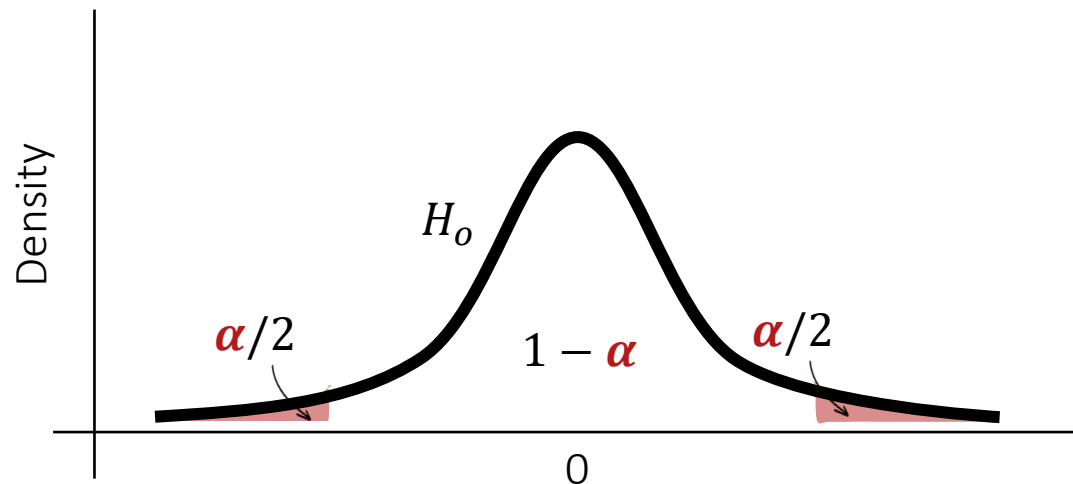
Histograms

Practice Time

Post-Hoc Tests, Multiple Comparisons, and P-Value Adjustment

ANOVA – In practice... Identifying groups responsible for significant differences

- We correct the H_0 rejection threshold for multiple testing, such as the overall risk of Type I Error remains α
- With the Bonferroni correction, we simply divide by the number of pair-wise comparisons

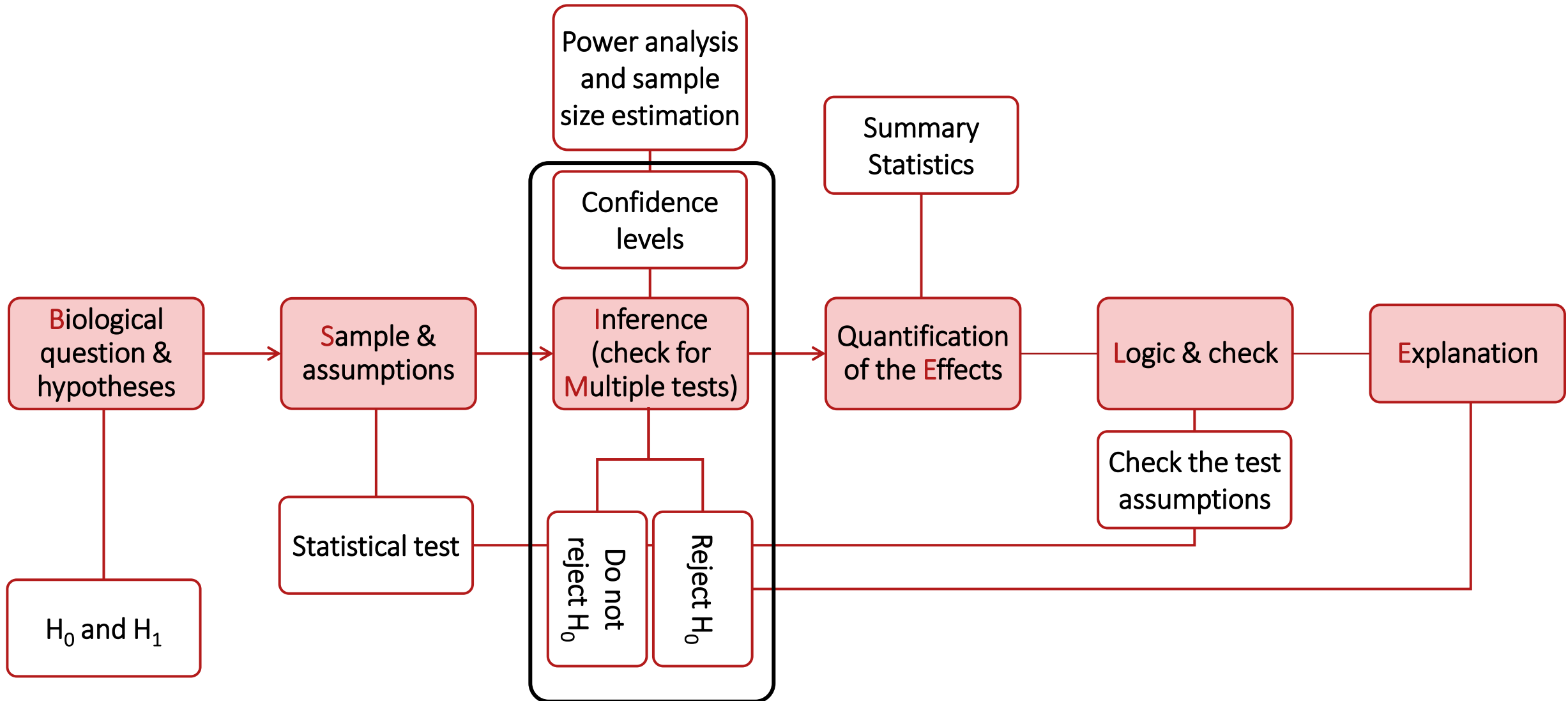


ANOVA – In practice... Identifying groups responsible for significant differences

- Pair-wise comparison but watch out for multiple testing and overall increase in Type I Error

Null hypothesis (H_0) is...	True ($\mu_1 = \mu_2$)	False ($\mu_1 \neq \mu_2$)
Reject	Type I Error False positive Probability = α = significance	Correct decision True positive (association) Probability = $1 - \beta$ = power
Fail to reject	Correct decision True negative (no association) Probability = $1 - \alpha$ = confidence	Type II Error False negative Probability = β

Hypothesis Testing Workflow



ANOVA – In practice... Identifying groups responsible for significant differences

- Different post-hoc pairwise tests...

Multiple Testing

- Applies in any context with repeated tests:
 - Pair-wise comparisons when > 2 groups
 - Consideration of multiple variables
 - E.g., body condition measured as weight, hip circumference, BMI, % fat, cholesterol level... (~ try all possible ways to assess body condition until one happens to show a significant association with the underlying factor of interest)

ANOVA – In practice... Identifying groups responsible for significant differences

- Different post-hoc pairwise tests...
 - Small number of pair-wise comparisons (i.e., small number of groups): **Bonferroni**
 - But reduces power, so if we want to favor power, we might prefer: **Šidàk**
 - Large number of pair-wise comparisons: **Tukey**...
 - ... but requires equal sample size across groups, if not: **Tukey-Kramer**
 - But they favor power so we might prefer: **Scheffe**
 - ...

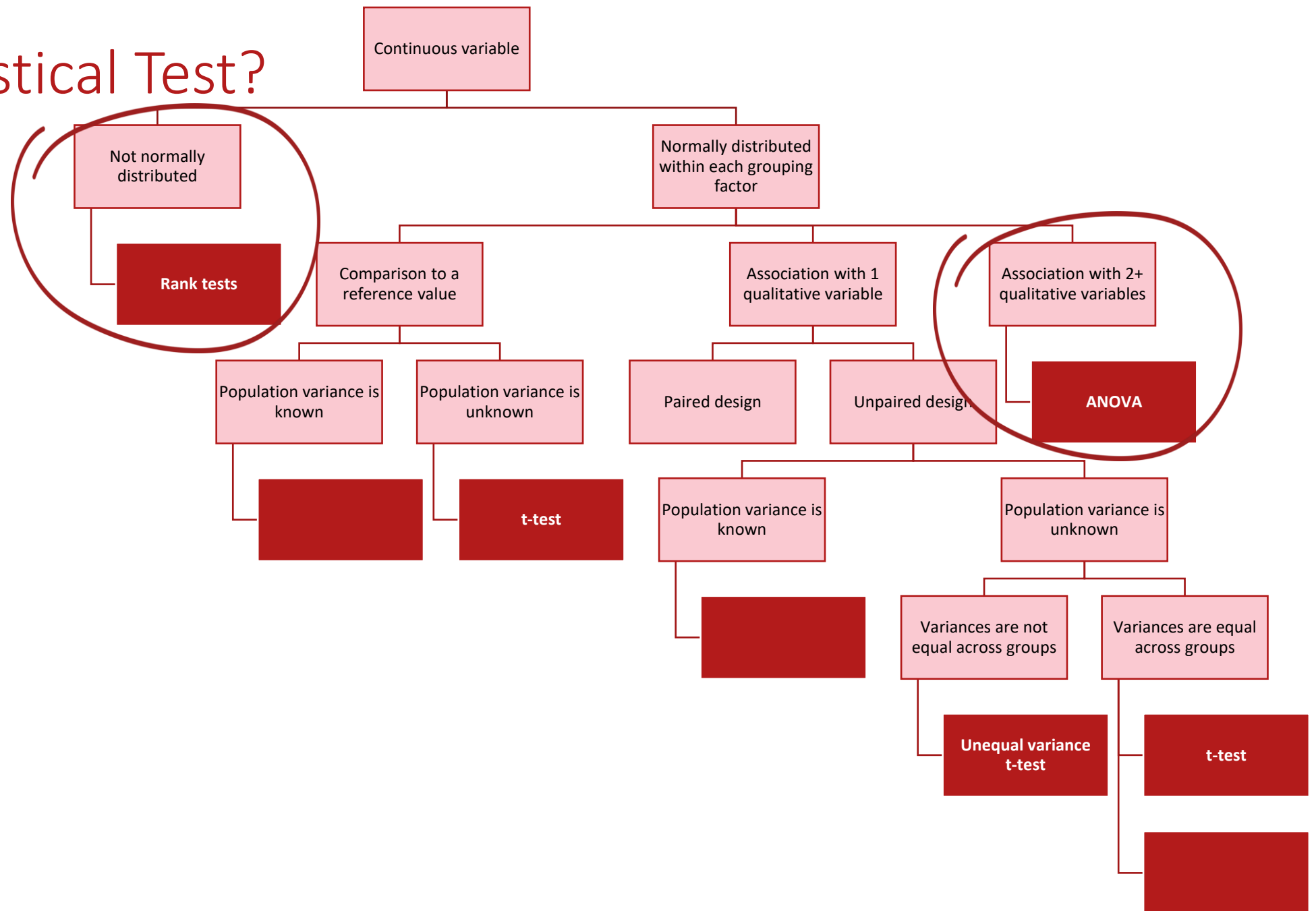
Practice Time

Non-Parametric Equivalent

What To Do When the Assumptions are Violated?

- Tests like the t-test that only apply when the data match a specific distribution are called **parametric tests**
- When this assumption is violated, we can use **non-parametric tests**
- Why do we bother with parametric tests? They have more statistical power (they are better at detecting associations that exist)
- Process is otherwise relatively similar

Which Statistical Test?



Which Statistical Test?

Context	Parametric Test	Non-Parametric Test
Comparison of 2 means	T-test	Wilcoxon rank test
Comparison of > 2 means	ANOVA	Kruskal-Wallis rank test

Practice Time

When There Is More Than One Grouping Factor

Two-way ANOVAs

Before Next Time (Friday February 27, in LH4)

- Practice: [Homework #05](#) due by **Thursday 5 PM**
- Use the [Guiding Quiz #05](#) if needed
Some of you should have received messages
- Check you are ready for next class: [Knowledge Check #05](#) due by **Friday 11 AM**

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
 - Come to office hours
 - **Tuesdays, 09:00-10:00 AM** (Jordan, Maddy) – Zoom possible
 - **Wednesdays, 05:00-06:00 PM** (Mainul, Princal, Daniel and Harsh) – Zoom possible
- If you cannot make it at these times, please contact the course assistant of your choice by email
- In S1-122, unless indicated otherwise on Canvas